B_c Meson Masses

To my mum, Hannah and Dad, Bright (RIP)

ACKNOWLEDGEMENT

First and foremost, we are immensely grateful to the creator of all things, The Lord God Almighty to whom all honour and glory be unto for His unconditional love and faithfulness which has brought me this far.

I say a special thank you to my supervisor**, Prof. Christine Davies** for her admirable patience, tolerance and guidance towards the success of this project, and dissertation. To all the Lecturers of School of Physics and Astronomy, I say thank you for preparing me with the skills to carry out this research with ease.

Lastly, I would be remiss in not mentioning my family, especially my mum. Their belief in me has kept my spirits and motivation high during this process.

ABSTRACT

Only one known meson, the B_c meson, is made up of two heavy quarks with different flavours: a charm and a bottom antiquark. This project aims to calculate the spectrum of B_c meson. As a result of the large size of the c and b quarks, the interactions of the charmonium and bottomonium mesons are thoroughly explored in the non-relativistic potential model with the purpose of determining the best fit potential model to estimate the mass and investigate the properties of the $c\bar{b}$ system. In order to do this, Numerov's method is used to numerically solve the three-dimensional radial Schrodinger wave equation in spherical polar coordinates using Power-law, Logarithmic, and Cornell potentials as the quark-antiquark interaction potentials. Since the potential models are spin-independent, the obtained energy eigenvalues are used to calculate the spin average mass spectra of the $b\bar{b}$, and $c\bar{c}$ systems in their ground state, radial excitation, and orbital excitation states, i.e., (1S, 2S, 3S, 1P, 2P), respectively. The results are then compared with the experimental spin-average values and other theoretical predictions of mass spectra. While the Cornell potential's mass spectrum values are high, the Martin and logarithmic potential models' values agreed with the experimental results. The Martin Potential model is discovered to be the best fit potential model to analyse the $c\bar{b}$ system after more testing was done to determine the energy splitting between the various energy levels. Every potential model applied in this project is also used in $c\bar{b}$ system, despite a stronger emphasis on the Martin potential model. After the energy splitting test was completed, the outcomes are consistent with the preceding theoretical predictions of the $b\bar{b}$ and $c\bar{c}$ systems which proves the non-relativistic nature of the bottom and charm quarks. In conclusion, the spin-average mass of B_c is calculated for 1S,2S,3S,1P,2P states respectively, and the pseudoscalar and vector meson (B_c^*) masses of the $c\bar{b}$ in ground state, and radial excitation state are calculated using the hyperfine splitting findings, and the outcomes are in good agreement with other theoretical predictions.

Table of Contents

ACKNOWLEDGEMENT ..... **ii**
ABSTRACT ..... **iii**
1 INTRODUCTION **1**
2 THEORETICAL BACKGROUND **4**
2.1 The Standard Model **4**
2.2 Quarks **4**
2.3 Quantum Chromodynamics (QCD) **5**
2.3.1 Properties of QCD **5**
2.4 Non-Relativistic Treatment **6**
2.5 Potential Models **6**
2.5.1 QCD-Motivated Models **7**
2.5.2 Non-QCD like Models **8**
2.5.3 Spin – independent potentials **8**
2.5.4 Cornell Potential **9**
2.5.5 Logarithmic Potential **9**
2.5.6 Martin (Power-Law) Potential **9**
2.6 Bottomonium (bb) **10**
2.7 Charmonium (cc) **10**
2.8 Bc Meson (cb) **11**
3 METHODOLOGY **13**
3.1 Three-Dimensional Schrodinger Equation **13**
3.2 Numerov Method **15**
3.3 Estimation of the Energy Eigenvalues of the Quarkonia **17**
4 RESULTS AND DISCUSSION **20**
4.1 The Energy Eigenvalues and Radial Wave Function plots **20**
4.2 Spin Average **23**
4.3 Mass Spectra of Bottomonium and Charmonium **23**
4.3 Hyperfine Splitting **27**
4.4 Mass Spectra of the Bc Meson **28**
5 CONCLUSION AND SUMMARY **32**
6 REFERENCES **34**
7 APPENDIX **37**
APPENDIX A: Code **37**
APPENDIX B: Eichten and Quigg result **39**
APPENDIX C: Gerstein et al result **41**

1 INTRODUCTION

The development of particle physics, which started with the discovery of atoms as the fundamental building blocks of chemical compounds, has continuously revealed new levels in the substructure of matter. And has convinced us that, at the present resolution limitations, all matter is made up of point-like spin 1/2 quarks and leptons, as indicated in table 1 below.

Table 1: The six quarks with their masses from the particle data group (PDG) (Workman et al., 2022) and electric charges

Quarks	Masses [GeV/c^2]	Electric Charges [e]
Top (t)	172.9	2/3
Bottom (b)	4.18	-1/3
Charm (c)	1.27	2/3
Strange (s)	0.093	-1/3
Down (d)	0.00467	2/3
Up (u)	0.00216	-1/3

Recently, there has been a resurgence in interest in the spectroscopy of the heavy flavoured hadrons because of the numerous experiments carried out by experimental facilities like Belle, ATLAS, BES, LHCb, etc. that have been continuously providing more accurate and new information about the hadrons from light flavour to heavy flavour sectors. (Workman et al., 2022). For the quantitative test of quantum chromodynamics (QCD) theory and the standard model, an understanding of heavy quarkonium systems and their characteristics is helpful (Kuchin & Maksimenko, 2013). These heavy quarkonium systems are ones in which at least one quark or antiquark is a member of a heavy flavour sector, especially charm or bottom. They are represented as $q\bar{q}$ mesonic systems, which contain the quarkonia (Charmonium, $c\bar{c}$ and bottomonium, $b\bar{b}$) and B_c ($b\bar{c}$ or by $c\bar{b}$) mesons. Toponium does not exist due to the top quark's enormous mass since it decays through the electroweak interaction before a bound state can arise. Typically, the term "quarkonium" only applies to charmonium and bottomonium and not to any of the lighter quark-antiquark states. With several narrow states below the threshold for open flavour generation, heavy quarkonia exhibit a rich spectroscopy (Faustov & Galkin, 2003). Many tests have been conducted on the $c\bar{c}$ and $b\bar{b}$ mesons, and the results have been encouraging. The anti-bottom, $\bar{b}$, and charm, c, quarks are two distinctive heavy-flavour quarks, and together they make up the Bc meson family, making it distinct from other meson families in the Standard Model. According to QCD theory, the $\bar{b}$ and c quarks are tightly bound in a compact system, with rich spectroscopy of excited states. Our knowledge of the strong interaction can be enhanced by

studying the heavy-quark dynamics in the B_C mass spectrum. The mass spectrum of the B_c states may be predicted with significantly higher accuracy than many other hadronic systems due to the existence of two heavy-flavour quarks (Bernet et al., 2020).

Important details regarding the quark-antiquark interactions may be learned from how well theoretical model predictions match up with observations. Such knowledge is extremely valuable since it is not feasible to derive the quark and antiquark potential from the fundamentals of quantum chromodynamics at the hadronic scale. Non-perturbative factors related to the intricate structure of the QCD vacuum must be taken into account at this scale. Due to all of this, the quark and antiquark potential at large and intermediate distances is theoretically uncertain. Most of the fundamental hadron resonances are created exclusively in this area of large and intermediate distances. Few models have been able to accurately predict the hadronic characteristics, from mass spectra to decay widths, despite several attempts or techniques to explain the hadron properties based on its quark structure. (Rai et al., 2008). The nonrelativistic potential models such as Buchmuller and Tye (Buchmuller & Tye, 1981), Martin (Martin, 1980), Logarithmic (Quigg & Rosner, 1977), Cornell (Eichten et al., 1978), Richardson (Richardson,1979), etc., were successful at the heavy flavour sectors for the mass prediction. These potential models are based on the idea that the interaction between the heavy quark and antiquark inside the quarkonium may be characterised by a central potential, and can be used to analyse the dynamics of heavy quarkonium in a non-relativistic manner. Due to the large masses of bottom and charm, and the low velocity of bottom and charm respectively, the quark-antiquark pair system can be treated non-relativistically by solving the three-dimensional Schrodinger equation to obtain the energy eigenvalues to discover the properties of the bound state. For the charmonium (ψ) and bottomonium (Υ) families, the non-relativistic potential model has proved effective, but the relativistic techniques perform better in the lighter sector. In reasonable agreement with the results of the experiments, certain prospective models have also made predictions about the masses and different decays of the heavy-heavy mesons (Abu-Shady & Ezz-Alarab 2019).

Extension of prior phenomenological potential model studies on quarkonium masses to their predictions of decay widths using nonperturbative techniques like nonrelativistic QCD (NRQCD) is necessary given the new importance of the heavy flavour investigations as the testing ground for the nonperturbative features. The NRQCD formalism made a great effort in this situation (Bodwin et al., 1995). For the heavy quark and antiquark, it consists of a non-relativistic Schrodinger field theory. The Schrödinger equation is crucial for explaining a

variety of events, including those in high energy physics (Abu-Shady & Ezz-Alarab, 2019). As a result, it's crucial to know the Schrodinger equation's solutions in order to determine the quarkonia's mass and other characteristics about it. The Numerov method will be employed with certain potentials to solve the Schrödinger equation's approximative solutions.

By solving the 3-dimensional Schrodinger equation to obtain the energy eigenvalues and corresponding wave functions of the heavy quarks using the Numerov method, this project aims to study the heavy-heavy flavour mesons in the charm and bottom sector within a conceptual approach of the Cornell potential, Power-law potential, and Logarithmic Potential models. It then applies the results to calculate the heavy quarkonia masses and study some of their properties. This project is computational, and the Schrodinger equation is solved using the Python programming language to code the Numerov method. A thorough analysis is provided when the data and graphs are compared to those found in the literature.

The structure of this essay is as follows: The theoretical background of this project is covered in Section 2 of this report, where a brief discussion of the standard model of particle physics is discussed. Additionally, a brief discussion of QCD, quarkonia, and the potentials of the non-relativistic quark model are presented.

The methods used to complete this project are described in Section 3 in more detail. It describes how the Numerov method was used to numerically solve the radial Schrodinger equation using the Python programming language. The gathered data is next analysed to create the relevant graphs.

The results attained and those found in the body of prior research are presented in Section 4. The results are then provided with an in-depth analysis and comparison. The project's conclusion is provided in Section 5, which also describes the project's potential future extensions as well as the work that has already been completed.

2 THEORETICAL BACKGROUND

2.1 The Standard Model

It has been discovered that everything in the universe is made up of a small number of fundamental building pieces known as fundamental particles, which are regulated by four fundamental forces, thanks to the theories and research of thousands of physicists since the 1930s. The Standard Model of particle physics integrates our current knowledge of the interactions between these particles with three of the four forces namely Strong Nuclear force, Weak Nuclear force, and Electromagnetic Force. It was created in the early 1970s and has effectively predicted a wide range of events while accurately explaining nearly all experimental data. The Standard Model has developed through time and via numerous tests into a well-tested physics theory. It is the currently reigning theory of Particle Physics, and has antiparticles for each of the 12 constituent fermions. Figure 1 below depicts the six quarks and six leptons that make up the fermions. We examine the quarks of the standard model in this section.

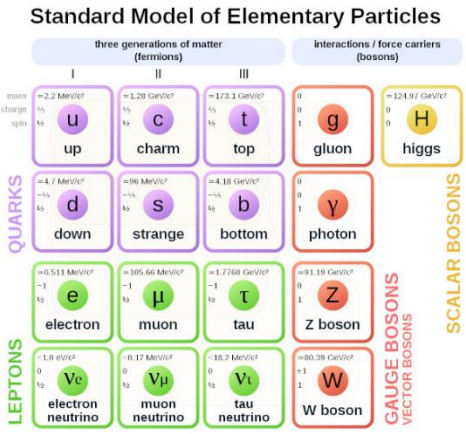


Figure 1: The standard model of elementary particles

2.2 Quarks

Quarks are fundamental subatomic particles that interact with other particles via the strong interaction. Figure 1 illustrates the six quark flavours that are different from one another in terms of mass and charge as shown in table 1, and they are top, bottom, charm, strange, down, and up. Quarks have no visible structure and cannot be divided into smaller particles, making them seem to be actual elementary particles. Quarks also appear to combine with one another or with antiquarks, which are their antiparticles, to create all hadrons. These hadrons have no colour charge, and it includes both baryons and mesons. The strong interactions within the hadrons can be described by the Quantum Chromodynamics (QCD). In this project, we will only work with the mesons, to be precise the heavy quark mesons.

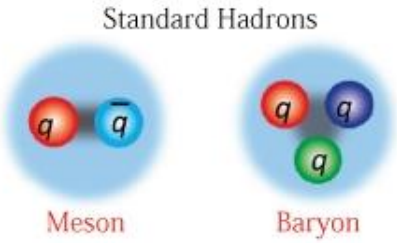


Figure 2: The Standard Hadrons

2.3 Quantum Chromodynamics (QCD)

QCD is the Standard Model theory used to describe the strong force interactions of the six quarks flavours that transform under the fundamental representation of the Special Unitary group of degree 3, SU(3) . It is described by a gauge theory. Gauge theories are those theories which are invariant under local gauge transformations, and are used in the standard model to describe the electromagnetic, weak and strong forces.

The Lagrangian of QCD is given as:

$$\mathcal{L}_{QCD} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} + \sum_{\{q\}}\bar{q}\left(i\gamma^\mu D_\mu - m_q\right)q \tag{1}$$

(Peskin, 2018)

Where {q} are the six quark flavours i.e., u, d, s, c, b, t. The field strength tensor in the Lagrangian is given by equation (2), in which the f^{abc} the SU(3) structure constants.

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu \tag{2}$$

$D_\mu = \partial_\mu - iT^a A^a_\mu$, T^a form a basis of the fundamental representation of the SU(3) algebra

2.3.1 Properties of QCD

The colour charge, a quantum number that the gluon fields themselves also carry, is used by the gluon fields to couple to the quarks. It is thought that the self-interaction that these fields add as a result of SU(3)'s non-abelian nature is related to the other unique characteristics of QCD. The physical spectrum only consists of colour singlet states; these states are either three quarks (baryons) or quark and antiquark pairs (mesons). We refer to this phenomenon as confinement.

Asymptotic freedom, or the fact that the QCD effective coupling constant $\alpha_s(q)$ decreases with decreasing distance scale (Fritzsch et al., 1973), is one of the key characteristics of QCD. At short distances, this means that the particles are virtually free. Additionally, this

characteristic of QCD prevents the internal energy scales of the hadrons we want to explore from being studied at high energies, leaving only perturbation theory applicable. But at low energies, it creates a mass gap commonly referred as scale of the quark confinement, Λ_{QCD}, which mostly contributes to the masses of the most hadrons. Perturbation theory is inapplicable at scales q $\sim \Lambda_{QCD}$, $\alpha_s(q) \sim 1$, and mostly the studies are done using the non-perturbative methods. The best established so far is the lattice QCD (Brambilla et al., 2004).

Quarks are often classified into light and heavy quarks. The heavy quarks are those whose mass is significantly more than the mass gap (QCD scale), i.e., $m_q \gg \Lambda_{QCD}$, (c, b, t) , and the light quarks are those whose mass is significantly less than the mass gap, i.e., $m_q \ll \Lambda_{QCD}, (u, d, s)$

2.4 Non-Relativistic Treatment

Hadrons containing heavy quarks are a difficult problem from the perspective of the description of QCD in subsection 2.3.1. As a result, many models have been used to overcome these challenges at the cost of wild approximations that later lose contact with QCD. Since the relativistic dynamics may be addressed without the approximation in lattice gauge theories, this project implements non-relativistic treatment to the heavy quarkonium dynamics which is easier to use because of the large size, and compares the obtained findings to the Experimental data by particle data group (PDG).

2.5 Potential Models

Researchers have been able to create effective tools for the study of QCD dynamics in heavy quark interactions because of the small parameter, Λ_{QCD}/m_q. The effective heavy quark theory (EHQT), which has been successfully used to examine hadrons containing a single heavy quark, is one of these techniques (Gershtein et al., 1995). The phenomenological potential models will be the focus on in this project. These potential models, which are based on the idea that the interaction between the heavy quark and its antiquark inside the quarkonium may be characterised by a potential, and can be used to analyse the dynamics of heavy quarkonium in a non-relativistic manner. Due to the large masses of bottom and charm, i.e., $m_{c,b} \gg \Lambda_{QCD}$ and the low velocity, i.e. $v \ll 1$, of bottom and charm respectively, the quark - antiquark pair system can be treated non relativistically by solving the three-dimensional Schrodinger equation to obtain the energy eigenvalues to discover the properties of the bound state. The solving of the Schrodinger equation is discussed in more detail in section 3. There are many phenomenological potential models to study heavy quarkonium

physics, and these potentials can be classified as QCD-like Potential, and QCD-motivated Potentials. Also, it can be classified into Spin dependent and Spin-independent Potentials.

Before discussing the exact potentials models employed in this research, it will first be briefly addressed how potentials models are classified as either non-QCD like or QCD motivated.

2.5.1 QCD-Motivated Models

Two static colour sources that are separated by distance, r in a colour singlet state is the inspiration for this type of model. The static $q\bar{q}$ potential is given by the first order calculation for a single gluon interaction as

$$V(r) \sim \frac{4}{3}\frac{\alpha_s(r)}{r} \quad (3)$$

(Peskin, 2018)

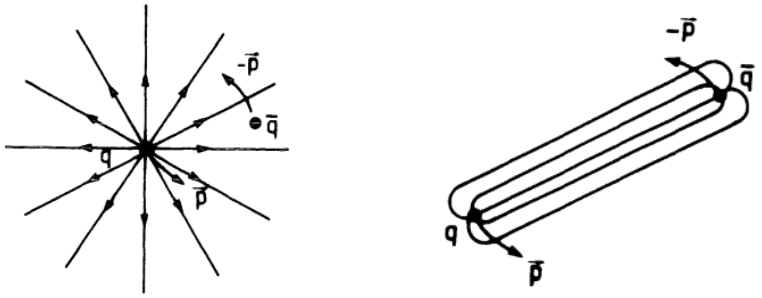
Where $\alpha_s(r)$ is strong coupling consatnt

Moreover, as can be seen in figure 3a below, this is equivalent to a static potential Coulomb field at a close range but with α_s rather than Quantum electrodynamics (QED) coupling. The colour lines of force extending between the two sources link together as a result of the gluon interactions at large separations, when the coupling becomes strong and perturbative theory is no longer applicable. Then, the static $q\bar{q}$ potential becomes

$$V(r) \sim \mathrm{Kr}, \quad \mathrm{r} \gg \Lambda \quad (4)$$

Where K is a proportionality constant, called the string tension.

This model just adds the two pieces together, i.e., Equation (3) and (4), on the potential's behaviour at the intermediate distance. The Cornell group (Eichten et al., 1978) was the group that initially presented this kind of model, and section 2.5.3 will go into further depth on this. Furthermore, calculations for the static $q\bar{q}$ potential at next to leading order as well as the running of the coupling constant are included in more complex models that have been put out. The Buchmuller and Tye potential, and the Richardson potential are two examples (Buchmuller & Tye, 1981), (Richardson, 1979).



(a) (b)

Figure 3: (a) Configuration of the $q\bar{q}$ potential's short-distance field demonstrating a Coulomb-type interaction driven by a single gluon exchange. **(b)**The large distance expected field configuration of the $q\bar{q}$ potential showing the linear increase of binding energy with separation (Peacock, 1986).

2.5.2 Non-QCD like Models

Even if one acknowledges the QCD-motivated models' effectiveness, the question of whether these accomplishments actually support theoretical predictions based on the QCD or only show that they are consistent with experimental data remains. The success of a class of logarithmic potentials created by Quigg and Rosner in 1977 (Quigg & Rosner, 1977) as well as the Martin potential (Martin, 1980), which bears little similarity to the asymptotic behaviour predicted by QCD theory at either short or large distances, helped alleviate these concerns later. Specifically, this will be covered in sections 2.5.4 and 2.5.5. These non-QCD-like models, which have been presented to empirically describe quarkonia spectroscopy, lack any theoretical underpinnings. This is generally done when the QCD-motivated potentials enable approximation in the region of the average distance between the heavy quarks (Gershtein et al., 1995). In addition to the potentials listed above, there are others that adopt the same approach. The logarithmic, power-law, and Cornell potential overlap in the range

$$0.51\ \mathrm{GeV}^{-1} < \mathrm{r} < 5.06\ \mathrm{GeV}^{-1} \qquad (6)$$

with an additive shift up to the energy scale, as shown in (Kiselev et al., 2001). Although these potentials have different asymptotic behaviour in the regions of very small r, and very large r.

2.5.3 Spin – independent potentials

We take into consideration of three spin-independent potentials models that provide a plausible explanation for the $c\bar{c}$, $b\bar{b}$, and $c\bar{b}$ spectra in this paper. These potentials are

Cornell Potential, Logarithmic Potential, and Martin Potential (Power-Law) and the parameters for the various potentials' models were given by Eichten and Quigg (Eichten & Quigg, 1994)

2.5.4 Cornell Potential

Equation 4 illustrates how a 1-gluon exchange-based central potential is produced using the perturbation theory for QCD. Due to the possibility of free quarks escaping, this coulomb potential cannot be the final solution (Davies, 1998). A second term is often assumed to be linear in *r* and to have the properties of a "string-like" confining potential in order to address this problem. Eichten and colleagues proposed this Cornell potential model in 1975 (Eichten et al., 1975). Therefore, this potential model has Coulomb-like behaviour at small distances, and the term confining the quarks rises linearly at large distances. Here is how the potential is expressed and be used in this project:

$$V_C(r) = \frac{4}{3}\frac{\alpha_s(r)}{r} + \frac{r}{a^2} \tag{7}$$

$$V_C(r) = \frac{\kappa}{r} + \frac{r}{a^2}$$

With $m_c = 1.84\,\frac{\text{GeV}}{\text{c}^2}$ $m_b = 5.18\,\frac{\text{GeV}}{\text{c}^2}$ $\kappa = \frac{4\,\alpha_s(r)}{3} = 0.52$ $a = 2.34\ \text{GeV}^{-1}$

2.5.5 Logarithmic Potential

Quigg and Rosner's non-relativistic heavy quark model with logarithmic binding from 1977 (Quigg & Rosner, 1977) has no connection to QCD; it is only proven to function. It was further motivated by the approximately mass-independent spacings between the $2\,^3S_1$ *and* $1\,^3S_1$ levels of charmonium (ψ) and bottomonium (ϒ) systems i.e.,

$$M(\Upsilon') - M(\Upsilon) \approx M(\psi') - M(\psi) \approx 0.59\ \text{GeV} \tag{8}$$

The logarithmic Potential is expressed as;

$$\text{V}_\text{L}(\text{r}) = -0.6635\ \text{GeV} + (0.733\ \text{GeV})\ \log\ (\text{r} \cdot 1\ \text{GeV}) \tag{9}$$

With $m_c = 1.5\ \text{GeV/c}^2$ $\text{m}_\text{b} = 4.906\ \text{GeV/c}^2$

2.5.6 Martin (Power-Law) Potential

Like the Logarithmic potential, this potential model likewise lacks QCD motivation. Because of their simple scaling characteristics, Quigg and Rosner advocated taking into account the potential of simple power type (Quigg &Rosner, 1997); however, Martin improved upon their idea in 1980 (Martin, 1980) by employing the following formula.

$$V(r) = A + Br^{\alpha} \tag{10}$$

A significant quantity of data was demonstrated as being understandable by carefully selecting the three parameters (A, B, α) and two quark masses. Therefore, the following is the expression for the Martin potential for this project (Eichten & Quigg, 1994)

$$V_M(r) = -8.064\ \text{GeV} + (6.898\ \text{GeV}) \log(\text{r} \cdot 1\ \text{GeV})^{0.1} \tag{11}$$

With $m_c = 1.8\ \text{GeV/c}^2$ $\quad \text{m}_\text{b} = 5.174\ \text{GeV/c}^2$

2.6 Bottomonium ($b\bar{b}$)

Since the bottomonium has the most observed states out of all the heavy quarkonium systems, we start with the $b\bar{b}$ system. The $b\bar{b}$ triplet states$\Upsilon(1S), \Upsilon(2S), \Upsilon(3S)$ were originally identified in copper and lead proton scattering by at the Fermilab in 1977 (Innes et al., 1977). The spin singlet state $\eta_b(1S)$, with a measured mass of 9.3889 $\pm$ 0.0027 GeV, was finally found by the Belle project in 2008 after several experimental tries (Aubert et al., 2008). Together, these experimental facilities BABAR (Lees et al. 2011), CLEO (Dobbs et al., 2012), and Belle (Mizuk et al., 2012), collaboration were able to observe the $\eta_b(2S)$. Radiative decays of the $\Upsilon(3S)$ and $\Upsilon(2S)$) led to the discovery of the two triplet P-wave states $\chi_{bj}(2P)$ and $\chi_{bj}(1P)$ with $J = 0,1,2$ in 1982 (Han et al., 1982) and 1983 (Klopfenstein et al., 1983), respectively. The continued operation of experimental facilities like Belle, ATLAS, BES, and LHCb is anticipated to increasingly yield accurate data. Due to its large mass, the $b\bar{b}$ system has a non-relativistic nature, allowing it to be treated as a two-body system composed of a heavy quark and antiquark. This makes an attempt to describe the dynamical properties of the $b\bar{b}$ system in terms of non-relativistic potential models plausible.

There are now 22 experimentally verified $b\bar{b}$ states (Workman et al., 2022). Therefore, it is crucial to investigate the masses of the $b\bar{b}$'s states from a theoretical standpoint. To verify the potential models, it is also necessary to test some of the decay channels in addition to doing computations on the mass spectra of each mesonic state.

2.7 Charmonium ($c\bar{c}$)

When two research teams at the Stanford Linear Accelerator Center (SLAC) and Brookhaven announced the detection of the new narrow resonance known as J/ψ with an observed mass of 3.1 GeV almost the same time, the charmonium system was found (Aubert et al., 1974). This was swiftly followed by the discovery of a new narrow state called ψ' with an observed mass of 3.105 $\pm$ 0.003 by the SLAC group (Augustin et al., 1974) and Lawrence Berkeley

Laboratory group (Abrams et al., 1974). These two resonances were understood to represent the bound states of a charm and its anticharm, the existence of which had been predicted years before. Our knowledge of hadron spectroscopy has been completely transformed by the discovery of the J/ψ and ψ' states, which showed that they could be accurately predicted by models that include qualitative QCD characteristics (Brambilla et al., 2004). It is possible to try a description of the dynamical characteristics of the $c\bar{c}$ system in terms of the non-relativistic potential models due to the heavy mass of the charm quark even though not at as heavy as bottom quark, which is approximately 1.27 $\mathrm{GeV/c^2}$ (see table 1)

2.8 B_c Meson ($c\bar{b}$)

A great deal of interest in $B_c$spectroscopy has been sparked by the Collider Detector at Fermilab (CDF) collaboration's discovery of the B_c meson with proton-antiproton collisions at $\sqrt{s}$ = 1.8 TeV, where s is the centre of mass energy, and with an observed mass of $M_{B_c} = 6.40 \pm 0.39 \pm 0.13$ (CDF collaboration, 1998).The result obtained by CDF collaboration for the B_c mass had a high uncertainty, henceforward a mass with a high accuracy was later seen in 2006 by CDF collaboration led by Abulencia and colleagues (Abulencia et al., 2006) This discovery showed the feasibility of experimental study of this system. A unique insight into the behaviour of heavy quarks is offered by the B_c mesons. The properties of B_c mesons are a peculiar case in quarkonium spectroscopy since they are the only quarkonia that include heavy quarks of different flavours, despite the fact that they are intermediate between $c\bar{c}$ and $b\bar{b}$ systems. They are more stable because they contain flavour and cannot disintegrate into gluons. Excited B_c states that are below the BD threshold, as shown in figure 5, can transition to the ground state pseudoscalar by radiative or hadronic transitions before weakly decaying.

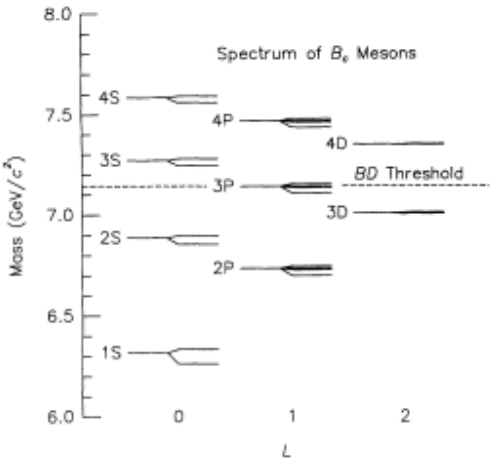


Figure 5: The B_c mass spectrum (Eichten & Quigg, 1994)

Despite the fact that there are less experimental data on the B_c system, its discovery has spurred fresh interest in the field of theory. Using the empirical mass formula and the logarithmic potential, Kwong and Rosner predicted the masses of the lowest vector and pseudoscalar states of the B_C system, with measured masses of $M_{B_C} = 6194 - 6292$ MeV and $M^*_{B_C} = 6284 - 6357$ MeV, respectively (Kwong & Rosner, 1991). Based on the QCD-motivated potentials i.e., Buchmuller and Tye, Cornell potential, and two non-QCD motivated potentials i.e., logarithmic and Martin Potentials, Eichten and Quigg estimated the masses and decays of the $c\bar{b}$, $b\bar{b}$ and $c\bar{c}$ systems (Eichten & Quigg, 1994). Using the QCD sum rule computations, Gershtein and colleagues also provided precise information on the masses and decay characteristics of the $c\bar{b}$, $b\bar{b}$ and $c\bar{c}$ systems (Gershtein et al., 1995).

Moreover, using a newly suggested potential form for quarkonia, Motyka and Zalewiski developed a nonrelativistic potential model to replicate the masses of the known $b\bar{b}$ spectrum within the experimental errors. By adding the common spin-dependent factors to the Hamiltonian, they furthered their research and derived the mass spectra of the $c\bar{c}$ and $c\bar{b}$ quarkonia as well as the leptonic decay constant and decay widths (Motyka & Zalewiski, 1998).

3 METHODOLOGY

3.1 Three-Dimensional Schrodinger Equation

In this project, the Schrodinger equation's numerical solution was used to estimate the energy eigenvalues of the quarkonia, as was explained in the sections 1 and 2. Since spherically symmetric systems, in which the Potential, V(r), is independent of the direction of r, are often best studied using spherical polar coordinates. The method was established in the spherical polar coordinate, as illustrated in figure 6. The coordinates in figure 6 was resolved to produce equation (12)

$$\left.\begin{aligned} x &= r sin\theta cos\phi \\ y &= r sin\theta sin\phi \\ z &= r cos\theta \end{aligned}\right\} \tag{12}$$

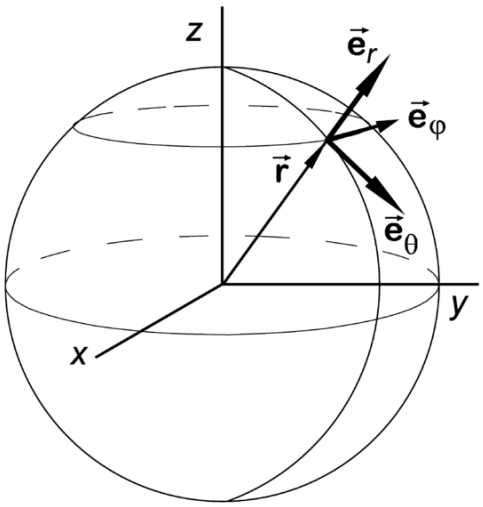


Figure 6: The spherical polar coordinate system

Starting the Schrodinger wave function equation,

$$-\frac{\hbar}{2m}\nabla^2\psi + V(r)\psi = E\psi \tag{13}$$

The ∇^2 in spherical polar coordinates is expressed as

$$\nabla^2 = \frac{1}{\boldsymbol{r}^2}\frac{\partial}{\partial r}\left(\boldsymbol{r}^2\frac{\partial}{\partial r}\right) + \frac{1}{\boldsymbol{r}^2\sin\theta}\frac{\partial}{\partial\theta}\left(sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\boldsymbol{r}^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \tag{14}$$

The Schrodinger equation in spherical coordinate is then given by:

$$-\frac{\hbar}{2m}\frac{1}{\boldsymbol{r}^2}\left[\frac{\partial}{\partial r}\left(\boldsymbol{r}^2\frac{\partial}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]\psi(r,\theta,\phi) + [V(r) - E]\psi(r,\theta,\phi) = 0 \tag{15}$$

Where r is the separation of $q\bar{q}$, V(r) = Potential energy, E =energy eigenvalue, $\psi =$ Schrodinger wave function, and m is the reduced mass of the quarks and antiquark in the bound state, and is given by

$$m = \frac{m_q m_{\bar{q}}}{m_q + m_{\bar{q}}} \tag{16}$$

For mathematical convenience, the variable separation approach was an efficient way for solving equation (15). In using the variable separation approach, it was presumed that the wave function $\psi(r, \theta, \phi)$ solution has a linear combination of functions that rely on spatial distance (r) and functions that depend on angles (θ, ϕ) as follows:

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi) \tag{17}$$

$R(r)$ is termed as the Radial wavefunction, $Y(\theta, \phi)$ is the spherical Harmonic function

Substituting equation (17) into equation (15) yields

$$\left[\frac{1}{R(r)}\frac{\partial}{\partial r}\left(\boldsymbol{r}^2\frac{\partial R(r)}{\partial r}\right) + \frac{2mr^2}{\hbar^2}[V(r) - E]\right] + \left[\frac{1}{Y(\theta,\phi)\sin\theta}\frac{\partial}{\partial\theta}\left(sin\theta\frac{\partial Y(\theta,\phi)}{\partial\theta}\right) + \frac{1}{\mathrm{Y}(\theta,\phi)\sin^2\theta}\frac{\partial^2 Y(\theta,\phi)}{\partial\phi^2}\right] = 0 \tag{18}$$

The first part of equation (18) simply depends on the variable r, whereas the second term depends on the angles (θ, ϕ), so they must separately be equal to a constant. $\ell(\ell + 1)$ was used as the separation constant, then equation (18) was separated into components that simply depend on r, as expressed in equation (19). ℓ is the orbital angular momentum quantum number

$$\frac{1}{R(r)}\frac{\partial}{\partial r}\left(\boldsymbol{r}^2\frac{\partial R(r)}{\partial r}\right) + \frac{2mr^2}{\hbar^2}[V(r) - E] = \ell(\ell + 1) \tag{19}$$

And the second term that depends on the angle (θ, ϕ) is

$$\frac{1}{Y(\theta,\phi)\sin\theta}\frac{\partial}{\partial\theta}\left(sin\theta\frac{\partial Y(\theta,\phi)}{\partial\theta}\right) + \frac{1}{\mathrm{Y}(\theta,\phi)\sin^2\theta}\frac{\partial^2 Y(\theta,\phi)}{\partial\phi^2} = -\ell(\ell + 1) \tag{20}$$

We can term equation (19) as the radial wave equation. A new variable as expressed in equation (21) was defined for convenience

$$\mathcal{U} = rR(r) \tag{21}$$

Substituting equation (21) into (19) yields

$$-\frac{\hbar}{2m}\frac{\partial^2\mathcal{U}}{\partial r^2} + \left[V(r) + \frac{\hbar^2}{2m}\frac{\ell(\ell + 1)}{r^2}\right]\mathcal{U}(r) = E\mathcal{U}(r) \tag{22}$$

Assuming the wave function in natural units, $\hbar = c = 1$, and multiplying (22) by $\frac{2m}{\hbar^2}$, it yields

$$\frac{\partial^2 \mathcal{U}}{\partial r^2} - 2m\left[V(r) - E_{n,l}\right]\mathcal{U}(r) - \frac{\ell(\ell+1)}{r^2}\mathcal{U}(r) = 0 \tag{23}$$

From (21), it can be seen that, the radial wave equation is identical to the Schrodinger equation in one dimension but with a modified effective potential as

$$\mathcal{V}_{eff} = V(r) + \frac{\ell(\ell+1)}{2mr^2} \tag{24}$$

This $\frac{\ell(\ell+1)}{2mr^2}$ is termed the "Centrifugal term

The Schrodinger wave function is normalised as:

$$\int_{-\infty}^{\infty} d^3\vec{r}\,|\psi_{nlm}(\vec{r})|^2 = 1 \tag{25}$$

So that the normalised radial wave function becomes

$$\int_{0}^{\infty} \vec{r}^2\, dr\,|R_{nl}(\vec{r})|^2 = 1 \tag{26}$$

The radial wave function at the origin was estimated after the equation (23) was normalised, and for $\ell = 0$, the value is just $R_{nl}^{\ell}(0)$ but when $\ell > 0$, equation (27) is used to determine the radial wave function at the origin since $R_{nl}^{\ell}(0) = 0$.

$$R_{nl}^{(\ell)}(0) = \left.\frac{d^{\ell} R_{nl}(r)}{dr^{\ell}}\right|_{r=0} \tag{27}$$

3.2 Numerov Method

Using the Numerov method, the Schrodinger's equation for a spherically symmetric potential was numerically solved in order to analyse the quarkonia states from a non-relativistic approach. The Numerov method is a numerical method used to solve second-order ordinary differential equations in which the first order is absent, as seen in equation (23). The Numerov method used to solve equation (23) was considered in the form;

$$\frac{d^2\mathcal{U}}{dr^2} = -g(r)\mathcal{U}(r) + s(r) \tag{28}$$

g(r) and s(r) are continuous functions on the domain $[r_{min}, r_{max}]$ The equation was solved as a boundary problem i.e., $\mathcal{U}(r_{min})$ and $\mathcal{U}(r_{max})$ were given. The interval was first discritised using equidistant points with step size, dr as illustrated in Figure 7.

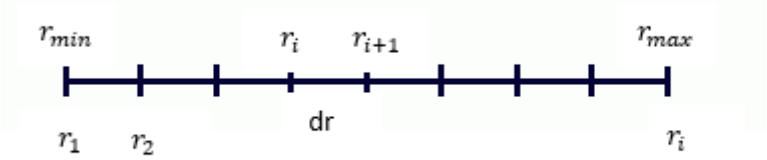


Figure 7: The discrete mesh of three-dimensional radial wave function

This mesh contains all the functions that are given. The second derivate employing the Taylor series was derived using the finite difference formula as follows

$$\mathcal{U}(r+dr) = \mathcal{U}(r) + dr\mathcal{U}'(r) + \frac{dr^2}{2}\mathcal{U}''(r) + \frac{dr^3}{12}\mathcal{U}'''(r) + \frac{dr^4}{24}\mathcal{U}''''(r) + \cdots \tag{29}$$

Similarly,

$$\mathcal{U}(r-dr) = \mathcal{U}(r) - dr\mathcal{U}'(r) + \frac{dr^2}{2}\mathcal{U}''(r) - \frac{dr^3}{12}\mathcal{U}'''(r) + \frac{dr^4}{24}\mathcal{U}''''(r) + \cdots \tag{30}$$

Adding equation (29) and (30) yields

$$\mathcal{U}(r+dr) + \mathcal{U}(r+dr) = 2\mathcal{U}(r) + dr^2\mathcal{U}''(r) + \frac{dr^4}{12}\mathcal{U}''''(r) + \mathcal{O}(dr^6) \tag{31}$$

Using the discrete notation, the second derivative becomes,

$$\mathcal{U}_i'' = \frac{\mathcal{U}_{i+1} + \mathcal{U}_{i-1} - 2\,\mathcal{U}_i}{dr^2} - \frac{dr^2}{12}\mathcal{U}'''' - \mathcal{O}(dr^4) \tag{32}$$

Hence the three-point central difference formula for the second derivative is

$$\mathcal{U}_i'' = \frac{\mathcal{U}_{i+1} + \mathcal{U}_{i-1} - 2\,\mathcal{U}_i}{dr^2} \tag{33}$$

Again, using the central difference formula, the fourth order derivative becomes

$$\mathcal{U}_i'''' = \frac{g_{i+1}\mathcal{U}_{i+1} + g_{i-1}\mathcal{U}_{i-1} - 2\,g_i\mathcal{U}_i}{dr^2} - \frac{s_{i+1} + s_{i-1} - 2\,s_i}{dr^2} - \mathcal{O}(dr^2) \tag{34}$$

Substituting equation (34) into (33) yields

$$\mathcal{U}_{i+1}\left(1+\frac{dr^2}{12}g_{i+1}\right) = 2\mathcal{U}_i\left(1-\frac{5dr^2}{12}g_i\right)-\mathcal{U}_{i-1}\left(1+\frac{dr^2}{12}g_{i-1}\right) + \frac{dr^2}{12}(s_{i+1}+10s_i+s_{i-1})+\mathcal{O}(dr^6) \tag{35}$$

Where $\mathcal{U}_i = \mathcal{U}(r_i)$, $g_i = g(r_i)$, $s_i = s(r_i)$, $dr = r_{i+1} - r_i$

i is the steps in dr away from the origin

This equation (35) is what is expected. As it can be seen, it is a three-term relation that connects the variables $\mathcal{U}_i$, $\mathcal{U}_{i+1}$ and $\mathcal{U}_{i-1}$, and it is efficient because the error is of the order of dr^6 which is one order smaller than the 4th order Runge Kutta, and most numerical methods that are of order of 4. The whole solution was constructed once we got two values of the solution U_i at succeeding mesh locations. These two necessary initial values were specified using the boundary conditions and asymptotic nature of the function, as discussed in section 3.3.

Comparing equation (35) and equation (23) resulted as:

$$\begin{aligned}\frac{d^2\mathcal{U}}{dr^2} &= 2m\left[V(r)_{eff}-E\right]\mathcal{U}_s \\ g(r) &= -2m\left[V(r)_{eff}-E\right] \\ s(r) &= 0\end{aligned} \tag{36}$$

Therefore, the Schrodinger equation was represented in complete set of i, where $i = 1,2,\ldots$, denoting the number of counts, $\ell = 0,1,$ representing the S and P waves respectively, and equation (35) was computed using python programming language to solve the radial wave function in order to estimate the energy eigenvalues of the $b\bar{b}$, $c\bar{c}$, and $c\bar{b}$ systems.

3.3 Estimation of the Energy Eigenvalues of the Quarkonia

To be able to regard the potentials as singular at the origin while estimating the energy eigenvalues, E_{nl} of equation (23), we took a guess for the energy and examined the wavefunction that resulted from applying the Numerov's method as seen in equation (36),

and we began in steps of r = dr + r_{min}, which was near to the origin. We then established the boundary conditions as

$$\mathcal{U}_1 = 0 \qquad \mathcal{U}_2 = dr \tag{37}$$

We observed the maximum of r r_{max}, such that $V(r) > E_{nl}$ for every $r > r_{max}$ in order to integrate the constraints of $R_{nl}(r)$ with an assumption that $V(r)$ does not oscillate at infinity and that r_{max} represents the turning point. The $R_{nl}(r)$, nodal structure was the next area of interest. It is widely known that the radial excitations have a limited number of nodes (q = 1, 2, 3,...) and that the ground state of the radial wave function (radial quantum number, n=0) has no nodes within $(0, \infty)$. Consequently, the following methodology was employed to estimate the E_{nl}. E_L, E_U which represent the lower and upper boundaries of E_{nl} (which is to be determined), were established as an interval for the energy variable, ϵ. After estimating the arithmetic mean $\epsilon_M = \frac{E_L + E_U}{2}$, equation (22) was integrated with $\epsilon = \epsilon_M$ starting from r = $r_{min} + dr$ with boundary constraints defined in (31). During the computation, these scenarios were possible.

- If the number of nodes surpassed q before the set r_{max} is reached, then further integration was halted since we knew, the $E_{nl} < \epsilon_m$ and changed the old E_U to the value of the ϵ_M that enhanced the upper bound to E_{nl}
- If the number of nodes remains smaller than q by reaching the set point of r_{max}. It was clear that a point for some $r > r_{max}$ have to reach since within classically forbidden region we know that $\mathcal{U}''(r)\mathcal{U}(r) > 0$. Then further integration was stopped. We therefore replaced the old E_L by ϵ_M, and continued the integration procedure with $r = dr + r_{min}$.

These integration computations for the quarkonia were done numerically by using the potential models from Eichten and Quigg (Eichten & Quigg, 1994), and the Numerov method described in section 3.2. If the bound of E_L and E_U are chosen badly, such that the energy$E_{nl} \notin (E_L, E_U)$, ϵ converges to the bound which lives next to the true E_{nl} as shown in figure 8.

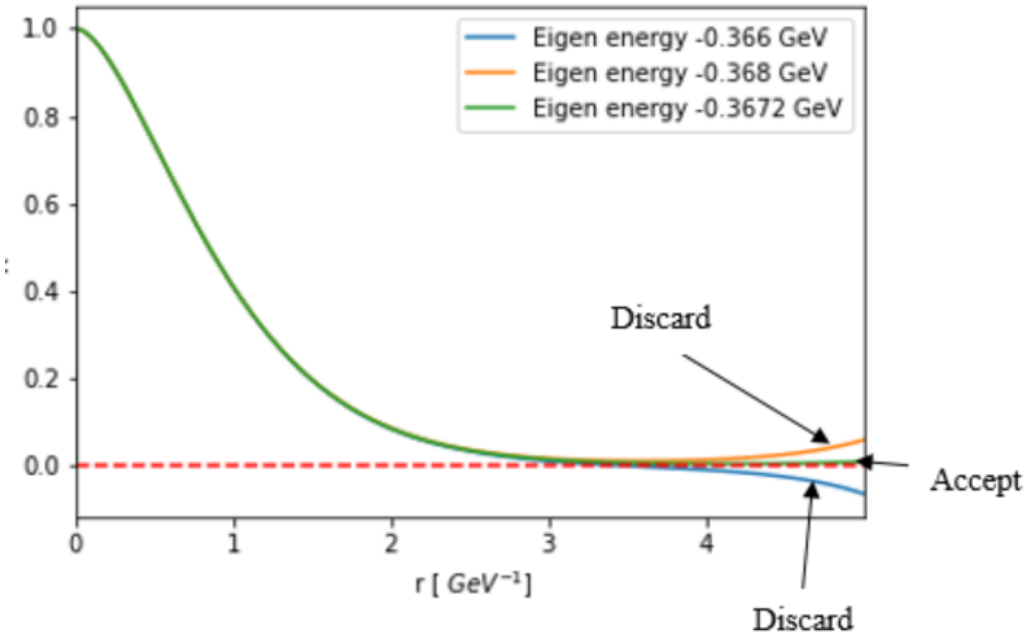


Figure 8: A plot showing the accepted and rejected energy eigenvalues of the radial wavefunction. The right answer for E_{nl} lies between the values of the orange and blue curves.

4 RESULTS AND DISCUSSION

4.1 The Energy Eigenvalues and Radial Wave Function plots

The energy eigenvalues needed to determine the masses for the $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems respectively, were obtained by solving the Schrodinger equation using the different potentials employed in this study. For the different potential models, the energy eigenvalues of the radial ground states, excitation states, and orbital excitation states, i.e., (1S,2S,3S,1P,2P), of the $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems are shown in tables 2, 3, and 4, respectively. Additionally, the plot of the radial wave function for the 1S, 2S, 3S, and 1P, 2P energy levels for the $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems, respectively. Figures 9a, 9b, and 9c illustrate how these plots agree with the expected nodal structure given in section 3.3.

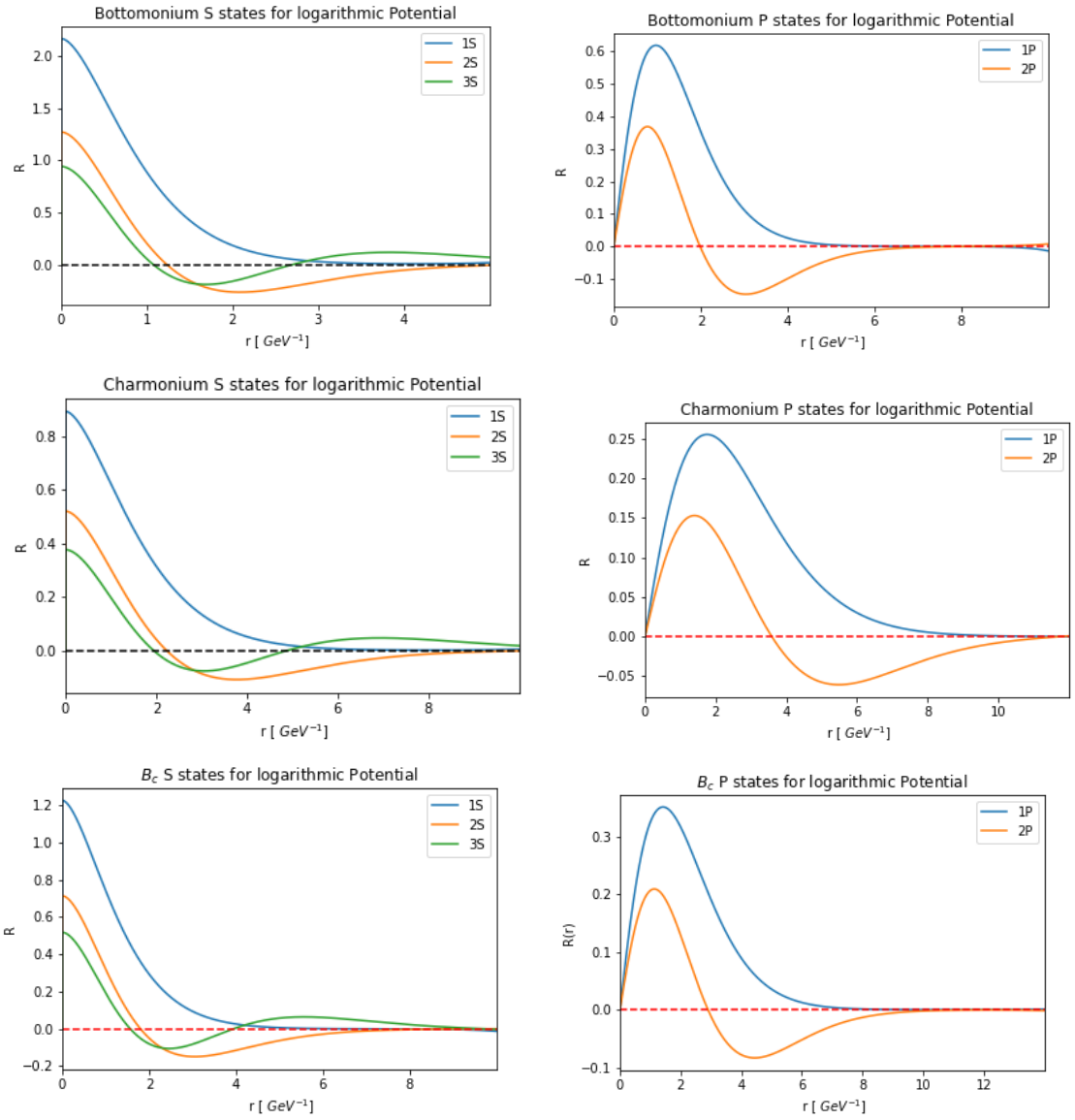


Figure 9a: The radial wave function plots of $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems for Logarithmic potential model depicting the nodal structure of 1S,2S,3S,1P,2P energy levels respectively

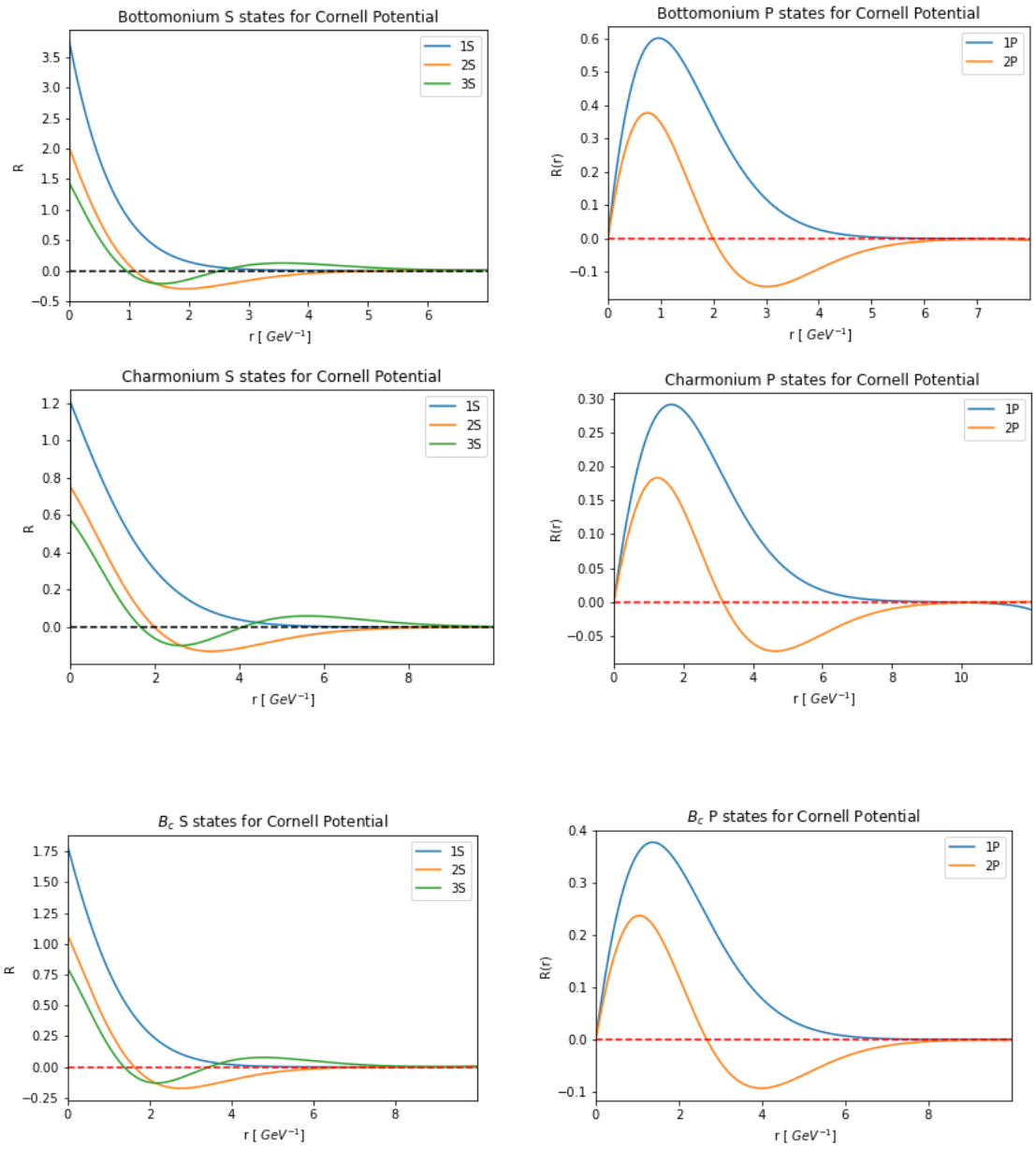
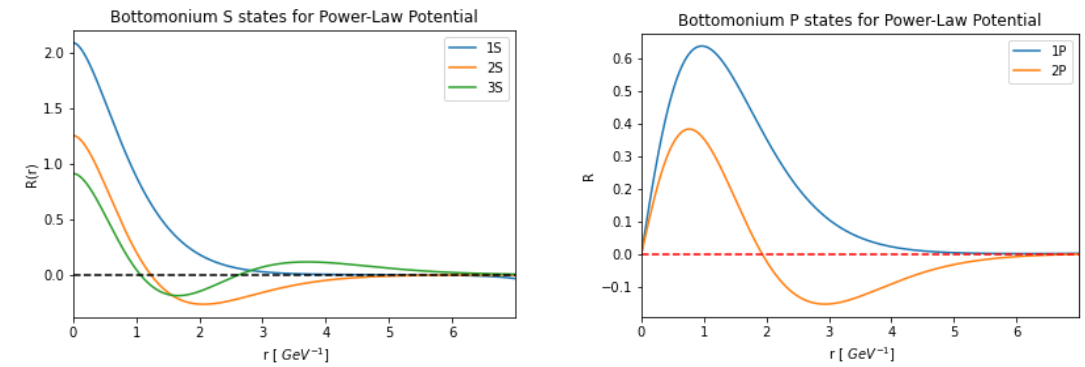


Figure 9b: The radial wave function plots of $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems for Cornell potential model depicting the nodal structure of 1S,2S,3S,1P,2P energy levels respectively



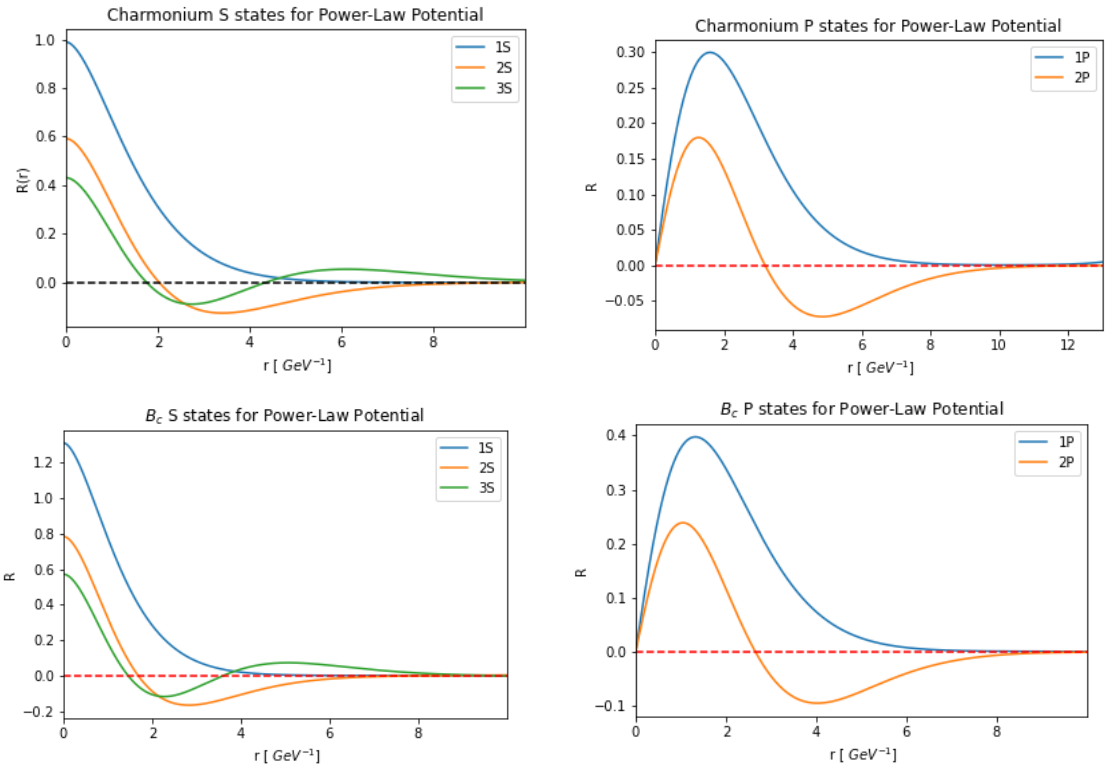


Figure 9c: The radial wave function plots of $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems for Martin(power-law) potentials depicting the nodal structure of 1S,2S,3S,1P,2P energy levels respectively

Table 2: The estimated Energy eigenvalues for the various energy levels of $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems for Power-law potential model in GeV

Level	Bottomonium	Charmonium	Bc
1S	-0.8742	-0.5035	-0.6445
2S	-0.2987	0.102	-0.0502
3S	0.0367	0.4542	0.2968
1P	-0.45848	-0.06637	-0.21518
2P	-0.07486	0.33696	0.1805

Table 3: The estimated Energy eigenvalues for the various energy levels of $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems for Logarithmic potential model in GeV

Level	Bottomonium	Charmonium	Bc
1S	-0.3672	0.0672	-0.0883
2S	0.2235	0.6563	0.4997
3S	0.5478	0.9807	0.8296
1P	0.0704	0.5047	0.3484
2P	0.4441	0.8797	0.7221

Table 4: The estimated Energy eigenvalues for the various energy levels of $c\bar{c}, b\bar{b}$ and $c\bar{b}$ systems for Cornell potential model in GeV

Level	Bottomonium	Charmonium	Bc
1S	-0.1705	0.2657	0.1116
2S	0.4215	0.8482	0.6813
3S	0.7658	1.2717	1.0686
1P	0.3279	0.6859	0.5468
2P	0.68198	1.1287	0.9491

4.2 Spin Average

As the potential models used in this project are spin independent. Therefore, our calculated results are the spin average masses. Therefore, we compared our results to the experimental spin average masses for the S and P states respectively using the relations

$$M(\bar{S}) = \frac{M(^1S_0) + 3M(^3S_1)}{4} \tag{38}$$

$$M(\bar{\chi}) = \frac{M(^1P_0) + 3M(^3P_1) + 5M(^3P_2)}{9} \tag{39}$$

The experimental spin average masses are computed from the experimental masses of the pseudoscalar and vector mesons.

4.3 Mass Spectra of Bottomonium and Charmonium

This section determined the mass spectra of the charmonium and bottomonium systems. Equation (40) was used to compute these heavy quarkonia masses.

$$M_{q\bar{q}} = m_q + m_{\bar{q}} + E_{nl} \tag{40}$$

Where $m_q, m_{\bar{q}}$ are the masses of each quark and antiquark composing of the quarkonium, E_{nl} is the estimated energy eigenvalue

Tables 5 and 6 compare the calculated masses of bottomonium and charmonium from this study's potential models with the experimental findings and other model predictions (see Appendix B and C). It is noted that only the Power-law and Logarithmic potential values, with a few MeV difference for the bottomonium and charmonium systems, respectively, are in good agreement with the experimental data out of the used potential models. This is due to the fact that calculations and parameters rely on the potential model, and as a result, Eichten and Quigg's (Eichten & Quigg, 1994) selection of mass parameters for the Cornell potential model was high.

Table 5: The spin-average mass spectra for the various energy levels of charmonium systems for Power-law, Logarithmic, and Cornell potential models in GeV

Level	Power-law	Logarithmic	Cornell	PDG
1S	3.0965	3.0672	3.9375	3.069 ± 0.00009
2S	3.7020	3.6563	4.5282	3.674± 0.00006
3S	4.0542	3.9897	4.9517	-
1P	3.5336	3.5047	4.3659	3.525 ± 0.00011
2P	3.9370	3.8797	4.8087	-

Table 6: The spin-average mass spectra for the various energy levels of bottomonium systems for Power-law, Logarithmic, and Cornell potential models in GeV

Level	Power-law	Logarithmic	Cornell	PDG
1S	9.474	9.445	10.190	9.445 ± 0.00023
2S	10.049	10.036	10.782	10.017 ± 0.00017
3S	10.385	10.360	11.126	10.355 ± 0.00025
1P	9.889	9.882	10.688	9.899 ± 0.0008
2P	10.273	10.256	11.0420	10.260 ± 0.0005

The potential models used in this work for the charmonium and bottomonium systems were further put to test by computing the energy splitting between the energy levels using the estimated energy eigenvalues, as shown in tables 7 and 8 correspondingly. Despite the Cornell potential's mass predictions not being in good agreement with the other potentials and experimental data, the energy splitting results are compared with the experimental data and other theoretical model predictions, and are in good agreement with all of them. Additionally, by comparing tables 7 and 8, it is shown that despite the $b\bar{b}$ and $c\bar{c}$ systems having different quark masses, the energy splitting for the charmonium and bottomonium systems roughly corresponds to one another, and it demonstrates that the non-relativistic nature of the systems

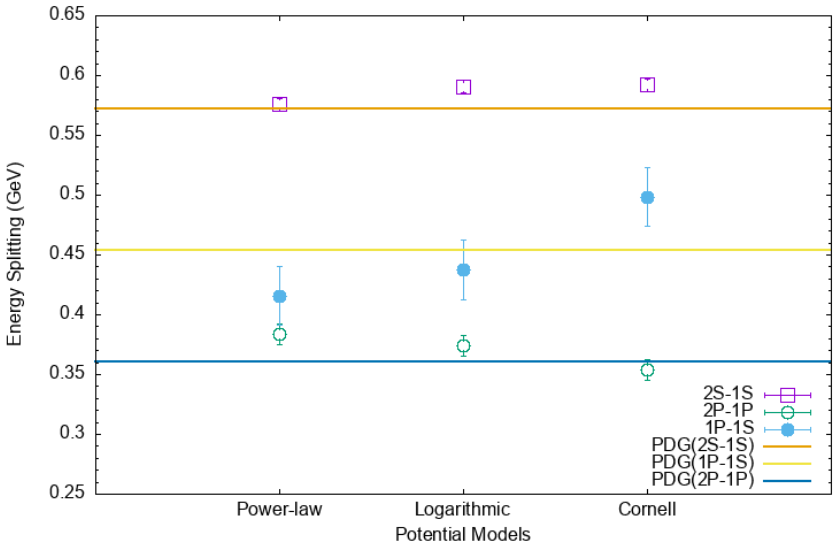
Table 7: The energy splitting between the various energy levels of bottomonium systems for Power-law, Logarithmic, and Cornell potential models in GeV with its uncertainty

	Power-law	Logarithmic	Cornell	PDG
3S-1S	0.9109	0.9150	0.9363	0.910 ± 0.00048
2S-1S	0.5755	0.5907	0.5920	0.572 ± 0.00040
1P-1S	0.4157	0.4376	0.4984	0.454 ± 0.00103
2P-1S	0.7993	0.8113	0.8525	0.815 ± 0.00073
2P-1P	0.3836	0.3737	0.3541	0.361 ± 0.00130

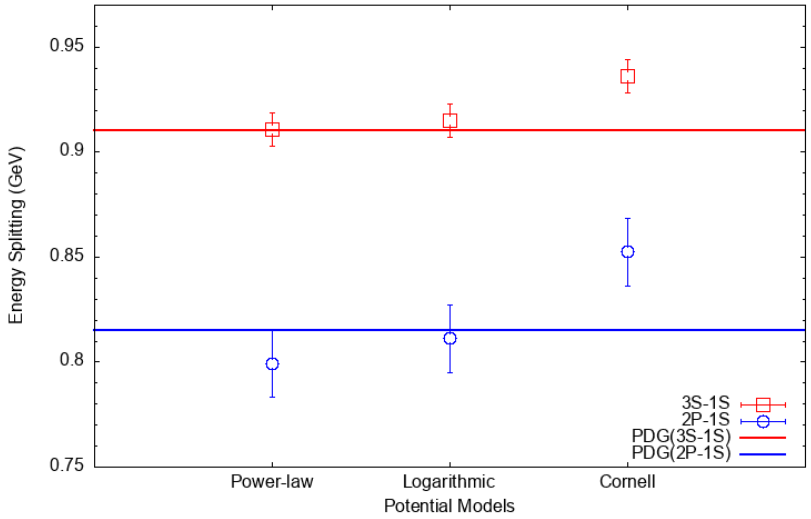
Table 8: The energy splitting between the various energy levels of charmonium systems for Power-law, Logarithmic, and Cornell potential models in GeV

	Power-law	Logarithmic	Cornell	PDG
3S-1S	0.9577	0.9135	1.0060	-
2S-1S	0.6055	0.5891	0.5825	0.605 ± 0.00015
1P-1S	0.4371	0.4375	0.4202	0.456 ± 0.0002
2P-1S	0.8405	0.8125	0.8630	-
2P-1P	0.4033	0.3750	0.4428	-

Figures 10 and 11 show plots of the energy splitting of the charmonium and bottomonium systems plotted against potential models used in this project. Since there is a lack of experimental data for other states, the theoretical predictions generated in this study are compared with other theoretical model predictions in appendices B and C, and they are shown to be in degree of fair agreement. The error bars represent the deviation that occurred in the potential models for the various energy splitting of the bottomonium and charmonium respectively.

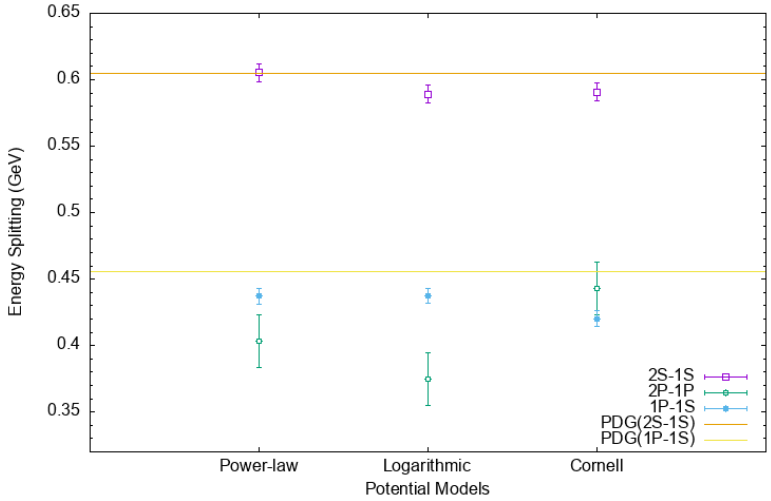


(a)

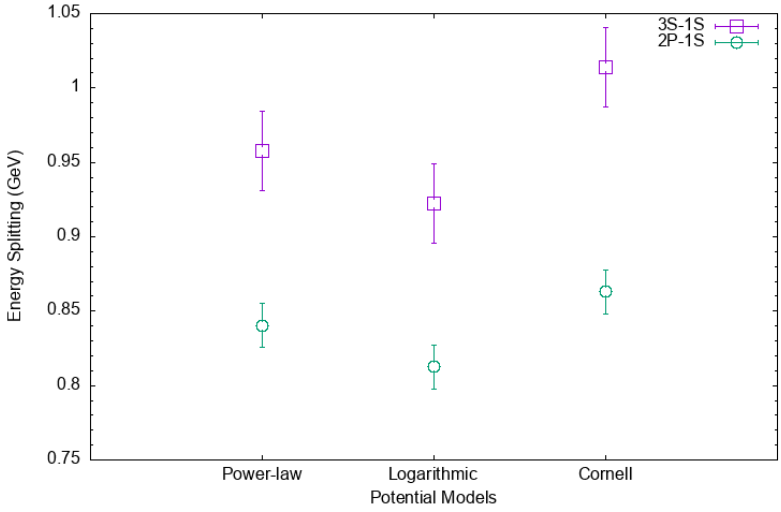


(b)

Figure 10: Energy splitting of bottomonium system for Power-law, Logarithmic, and Cornell potentials showing (a) 2S-1S, 2P-1P,1P-1S respectively and (b) 3S-1S, 2P-1S comparing with experimental values. The horizontal lines represent the experimental values taken from the PDG (Workman et al., 2022)



(a)



(b)

Figure 11: Energy splitting of charmonium system for Power-law, Logarithmic, and Cornell potentials showing (a) 2S-1S, 2P-1P,1P-1S respectively and (b) 3S-1S, 2P-1S. The horizontal lines represent the experimental values taken from the PDG (Workman et al., 2022)

4.3 Hyperfine Splitting

To determine whether a possible model is the best match, it must undergo a number of tests. The hyperfine splitting is one property that has been calculated in this section and compared to experimental results and other theoretical predictions. On each potential model, the hyperfine splitting, ΔM_{hyp}, the smallest change in energy levels, is computed as the difference between the vector and pseudoscalar group states as

$$M(^3S_1) - M(^1S_0) = \frac{32\pi\alpha_s|\psi(0)|^2}{9m_i m_j} = \frac{8\alpha_s|R(0)|^2}{9m_i m_j} \tag{41}$$

Where m_i, m_j are the masses of the quark used.

In order to calculate the mass difference between the vector and pseudoscalar mesons, equation (41) is used, because our calculations in this project is based on spin-independent potentials, and it does not allow the calculation of vector mesons. Instead, the hyperfine splitting is directly proportional to the radial wavefunction at origin, $\left|R_{nl}^{\ell}(0)\right|^2$ of the $b\bar{b}, c\bar{c}$ and $c\bar{b}$ systems, as shown in tables 9, 10, and 11 respectively for power-law, Logarithmic and Cornell potentials. Table 12 shows the results, which are in good agreement with the experimental values with only a few MeV differences of all the potentials models for $b\bar{b}$, $c\bar{c}$ systems, and other theoretical predictions for $c\bar{b}$ systems.

Table 9: Radial wave function at the origin for bottomonium system

Level	Power-law	Logarithmic	Cornell
1S	4.4242	4.6887	14.0469
2S	2.4592	2.4585	5.6885
3S	1.7703	1.6903	4.2306

Table 10: Radial wave function at the origin for charmonium system

Level	Power-law	Logarithmic	Cornell
1S	0.9785	0.7944	1.4574
2S	0.5465	0.4083	0.9296
3S	0.3901	0.2798	0.7892

Table 11: Radial wave function at the origin for B_c system

Level	Power-law	Logarithmic	Cornell
1S	1.7157	1.5178	3.2115
2S	0.9573	0.7702	1.7683
3S	0.6977	0.5666	1.4504

Table 12: The hyperfine splitting of ground-state and radial excitation masses of $c\bar{c}, b\bar{b}, c\bar{b}$ systems for Power-law, Logarithmic and Cornell Potentials in GeV

	Mesons	Power-law	Logarithmic	Cornell	PDG
1S	$M(J/\psi) - M(\eta_c)$	0.1154	0.1161	0.1186	0.113 ± 0.000070
	$M(\Upsilon) - M(\eta_b)$	0.0632	0.0641	0.1443	0.0623 ± 0.00032
	$M(B_c^*) - M(B_c)$	0.0704	0.0678	0.0928	-
2S	$M(J/\psi) - M(\eta_c)$	0.0645	0.0597	0.0757	0.0486 ±0.00116
	$M(\Upsilon) - M(\eta_b)$	0.0351	0.0336	0.0584	0.0243 ± 0.00381
	$M(B_c^*) - M(B_c)$	0.0393	0.0344	0.0511	-
3S	$M(J/\psi) - M(\eta_c)$	0.0460	0.0409	0.0757	-
	$M(\Upsilon) - M(\eta_b)$	0.0253	0.0231	0.0434	-
	$M(B_c^*) - M(B_c)$	0.0286	0.0253	0.0419	-

4.4 Mass Spectra of the B_c Meson

The Cornell, Logarithmic, and Martin (Power-law) Potentials, as well as other phenomenological non-relativistic potential models, have all been used to describe the B_c meson. Due to the fact that the families of Charmonium and Bottomonium spectroscopies

properties are thoroughly explored experimentally (Workman et al., 2022), the phenomenological potential models, that were employed in subsection 2.5.3 of this study were used to estimate the mass and other properties of this meson because this method of examination was justified as the results of the theoretical mass and energy splitting predictions of the bottomonium and charmonium are found to be in good agreement with the experimental values. Although the logarithmic and power-law potential tend to be in good agreement with the experimental values but the most effective model after the analysis of $b\bar{b}$ and $c\bar{c}$ system is the power-law potential. Hence, we focused more on the Power-law to perform calculations on the B_c meson's triplet and singlet states. The spin average mass of the $c\bar{b}$ system was calculated with expression in equation (42) even though, equation (40) would have calculated the masses. This is useful because the first term in the equation (42) removes the dependence of the quark masses in the potential models, and the results of the various energy levels for the potential models are presented in table 13 which proves the non-relativistic nature of the $c\bar{b}$ meson since the energies in all the potential models are approximately equal.

The mass of the $c\bar{b}$ system was calculated with this expression as follows

$$M_{B_c} = (E_{B_c} - \frac{1}{2}(E_{b\bar{b}} + E_{c\bar{c}})) + \frac{1}{2}(M_{b\bar{b}} + M_{c\bar{c}}) \tag{42}$$

Where $E_{B_c}, E_{b\bar{b}}, E_{c\bar{c}}$ are the estimated energy eigenvalues of $c\bar{b}, b\bar{b}, c\bar{c}$ systems respectively. The results obtained for the $c\bar{b}$ system is presented in table 9.

Table 13: The energies that eliminate the set masses of the bottom and charm for $c\bar{b}$ in the potentials in GeV

Level	power-law	logarithmic	Cornell
1S	0.04435	0.06170	0.06400
2S	0.04815	0.05980	0.04645
3S	0.05135	0.06535	0.04985
1P	0.04725	0.06085	0.03988
2P	0.04945	0.06020	0.04376

Table 14: The spin average mass spectra for the various energy levels of B_c system for Power-law, Logarithmic, and Cornell potential models in GeV

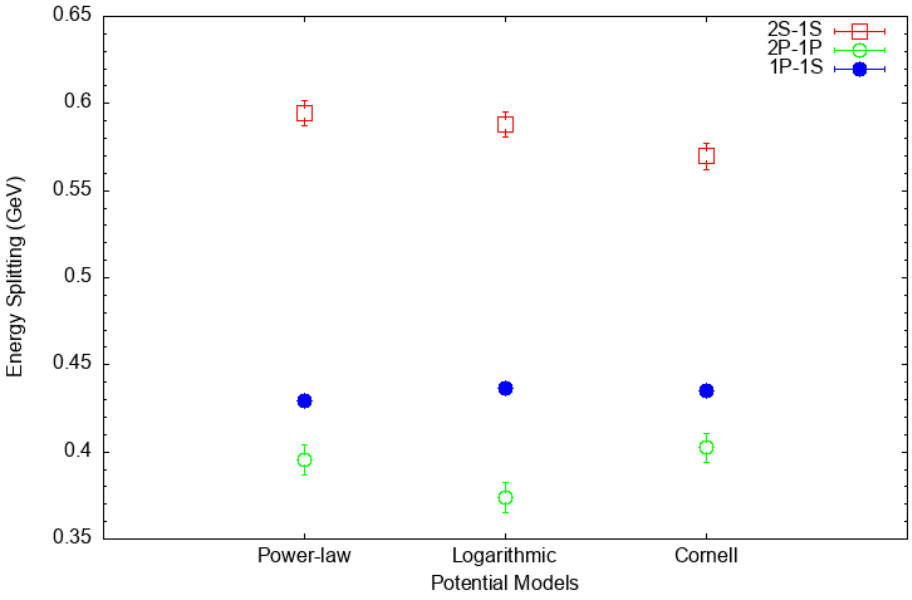
Level	Power-law	Logarithmic	Cornell	PDG
1S	6.3295	6.3177	7.1316	-
2S	6.9238	6.9057	7.7013	-
3S	7.2708	7.2356	8.0886	-
1P	6.7588	6.7544	7.5668	-
2P	7.1545	7.1281	7.9691	-

Although the Power-law and Logarithmic potentials mass predictions are in good agreement with the experimental values, i.e., 1S and 2S, and other theoretical mass predictions in appendix B and C, respectively, it can be seen in table 14 that the discrepancy that occurred in the Cornell Potential for the $b\bar{b}$ and $c\bar{c}$ systems reappeared in the $c\bar{b}$ system. This can support the claim that, there were uncertainties that accompanied the quark mass used by Eichten and Quigg (Eichten & Quigg, 1994). To test this further, the energy splitting of the $c\bar{b}$ system was estimated, and the results are shown in table 15.

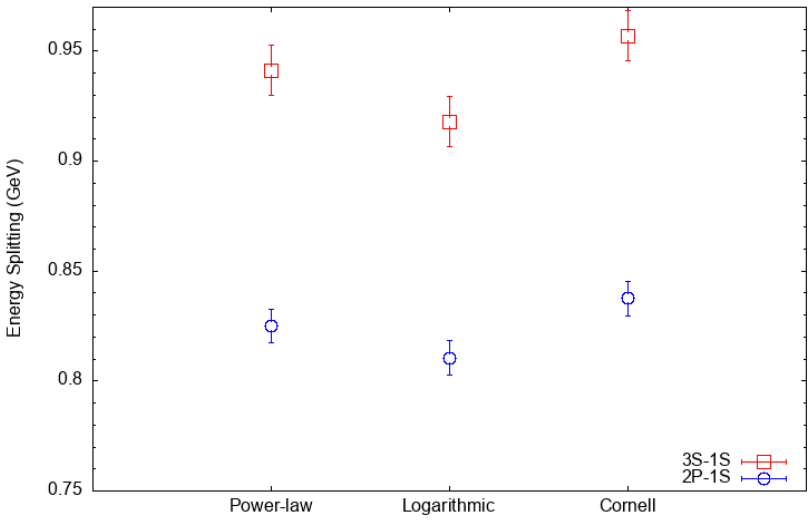
Table 15: The energy splitting between the various energy levels of B_c systems for Power-law, Logarithmic, and Cornell potential models in GeV

	Power-law	Logarithmic	Cornell	PDG
3S-1S	0.9413	0.9179	0.9570	-
2S-1S	0.5943	0.5880	0.5697	-
1P-1S	0.4293	0.4367	0.4352	-
2P-1S	0.8250	0.8104	0.8375	-
2P-1P	0.3957	0.3737	0.4023	-

Since there are only 2 verified experimental states of the $c\bar{b}$ system at this time, namely 1S and 2S (Workman et al., 2022), it was not possible to compare the energy splitting between different levels with experimental values but prospect is good that the predictions in this project will be in the experimental range when it is discovered in the future. However, the obtained results were in good agreement with those from Eichten and Quigg (Eichten & Quigg, 1994), and Kwong and Rosner (Kwong & Rosner, 1991). The projections for the energy splitting of the $c\bar{b}$ system, however, are approximately the same to predictions for the $b\bar{b}$ and $c\bar{c}$ systems, as seen in tables 7 and 8, respectively. The outcomes of the energy splitting between the energy levels are shown in Figure 12.



(a)



(b)

Figure 12: Energy splitting of B_C system for Power-law, Logarithmic, and Cornell potentials showing (a) 2S-1S, 2P-1P,1P-1S respectively and (b) 3S-1S, 2P-1S.

The vector, and pseudoscalar states of the B_c meson were computed using equation (43) after solving the hyperfine splitting of the bound states.

$$M(\bar{S}) = M_{triplet} - \frac{1}{4}(M_{triplet} - M_{singlet}) \tag{43}$$

Where $M_{triplet} - M_{singlet}$ is the hyperfine splitting. $M(\bar{S})$ is the spin average mass of the S states in table 13. As a result of an investigation of the $b\bar{b}$ and $c\bar{b}$ systems, the power-law potential model was found to be the most efficient model for the computations. The mass

predictions of the vector and pseudoscalar states for the 1S, 2S, and 3S states were computed using the power-law potential, the findings are shown in Table 15. The pseudoscalar findings for the 1S and 2S states are in good agreement with the experimental values. Even though the triplet state has not yet been identified, this project's mass predictions show strong agreement with other theoretical predictions (see Appendix B and C).

Table 15: The mass spectra of B_c's triplet state $(^3S_1)$ $[B_c^*]$ and singlet state $(^1S_0)$ $[B_c]$ for 1S, 2S, 3S using Power-law potential model in GeV

State	B_c	B_c^*	$PDG(^1S_0)$
1S	6.2767	6.3471	*6.2745 ± 0.00017*
2S	6.8944	6.9336	6.8712 ± 0.001
3S	7.2494	7.2780	-

5 CONCLUSION AND SUMMARY

The Numerov method was used in this project to solve the radial Schrodinger wave equation and determine the energy eigenvalues of the bottomonium, charmonium, and B_c mesons. Due to the large mass of the bottom and charm quarks, they were computed using the non-relativistic potential model's approach. The Martin, Logarithmic, and Cornell potentials are the models employed for the quark-antiquark interactions potential. Since the potential models were spin-independent, the obtained results for the energy eigenvalues were used to calculate the spin average mass spectra (1S, 2S, 3S, 1P, 2P) of bottomonium and charmonium. The results were then compared to the experimental values and other theoretically predicted mass spectra. When compared to the experimental data, the Martin and logarithmic potential models agreed, however, the Cornell potential values were high. The Martin (power-law) Potential model was determined to be the most appropriate potential model for our calculations given the nature of the error bar after the analysis. This was determined after conducting a further test to determine the energy splitting between the various energy levels to cancel many uncertainties that were present with the bare quark masses.

With the power-law potential serving as the most effective model among the potentials in this project, the vector and pseudoscalar of the B_c meson were calculated using the findings of hyperfine splitting. The results for the 1S, 2S, and 3S for the pseudoscalar state are, 6.2767 GeV, 6.8944 GeV, and 7.2494 GeV respectively. Since only the 1S and 2S states of the B_c

singlet have been identified experimentally, our result was compared, and it is in good agreement with an error difference of about 0.04% and 0.4%, respectively. Although the 3S has not yet been found experimentally, the outcome is much in line with other theoretical predictions. In good agreement with other theoretical predictions, unidentified triplet state of the B_c bound state, i.e., B_c^* for 1S, 2S, and 3S, were observed to have a mass of 6.3471 GeV, 6.9336 GeV, and 7.2780 GeV, respectively.

Despite the fact that the results of the non-relativistic approach used for this project are in good agreement with the experimental results and other theoretical predictions, the relativistic correction was not taken into consideration. The accuracy of the predictions would have been improved by the inclusion of relativistic correction. Alternatively, there are additional methods that provide more accurate predictions. Allison and colleagues used one such method in which they used ensembles of lattice gauge fields from the MILC collaboration that had a quark sea with two flavours (Allison et al., 2005). In conclusion, we anticipate that the predictions of the B_c^* state will fall within the range of the experimental results in the future.

6 REFERENCES

Abrams, G. S., Briggs, D., Chinowsky, W., Friedberg, C. E., Goldhaber, G., Hollebeek, R. J., ... & Vannucci, F. (1974). Discovery of a second narrow resonance in e+ e− annihilation. *Physical Review Letters*, *33*(24), 1453.

Abulencia, A., Acosta, D., Adelman, J., Affolder, T., Akimoto, T., Albrow, M. G., ... & Errede, S. (2006). Measurement of the B c+ Meson Lifetime Using the Decay Mode B c+→ J/ψ e+ ν e. *Physical review letters*, *97*(1), 012002.

Abu-Shady, M., & Ezz-Alarab, S. Y. (2019). Trigonometric Rosen–Morse potential as a quark–antiquark interaction potential for meson properties in the non-relativistic quark model using EAIM. *Few-Body Systems*, *60*(4), 1-11.

Aubert, J. J., Becker, U., Biggs, P. J., Burger, J., Chen, M., Everhart, G., ... & Lee, Y. Y. (1974). Experimental observation of a heavy particle J. *Physical Review Letters*, *33*(23), 1404.

Aubert, B., Bona, M., Karyotakis, Y., Lees, J. P., Poireau, V., Prencipe, E., ... & De Sangro, R. (BABAR) (2008). Observation of the Bottomonium Ground State in the Decay ϒ(3 S)→ γ η b. *Physical review letters*, *101*(7), 071801.

Augustin, J. E., Boyarski, A. M., Breidenbach, M., Bulos, F., Dakin, J. T., Feldman, G. J., ... & Zipse, J. E. (1974). Discovery of a Narrow Resonance in e+ e− Annihilation. *Physical Review Letters*, *33*(23), 1406.

Bernet, R., Müller, K., Owen, P., Serra, N., Steinkamp, O., & LHCb Collaboration. (2020). Precision measurement of the B_c^+ meson mass. *Journal of High Energy Physics*, *7*, 123.

Bodwin, G. T., Braaten, E., & Lepage, G. P. (1995). Rigorous QCD analysis of inclusive annihilation and production of heavy quarkonium. *Physical Review D*, *51*(3), 1125.

Brambilla, N., Krämer, M., Mussa, R., Vairo, A., Bali, G., Bodwin, G. T., ... & Zieminski, A. (2004). Heavy quarkonium physics. *arXiv preprint hep-ph/0412158*.

Buchmüller, W., & Tye, S. H. (1981). Quarkonia and quantum chromodynamics. *Physical Review D*, *24*(1), 132.

CDF Collaboration, F. Abe et al. (1998). Observation of B_c Mesons in p$\bar{p}$ Collisions at sqrt {s}= 1.8 TeV. *arXiv preprint hep-ex/9804014*.

Davies, C. (1998). The heavy hadron spectrum. In *Computing Particle Properties* (pp. 1-64). Springer, Berlin, Heidelberg.

Dobbs, S., Metreveli, Z., Tomaradze, A., Xiao, T., & Seth, K. K. (2012). Observation of the η b (2 S) Meson in ϒ (2 S)→ γ η_b (2 S), η_b (2 S)→ Hadrons and Confirmation of the η_b (1S) Meson. *Physical Review Letters*, *109*(8), 082001.

Ebert, D., Faustov, R. N., & Galkin, V. O. (2003). Properties of heavy quarkonia and Bc mesons in the relativistic quark model. *Physical Review D*, *67*(1), 014027.

Eichten, E., & Feinberg, F. (1981). Spin-dependent forces in quantum chromodynamics. *Physical Review D*, *23*(11), 2724.

Eichten, E., Godfrey, S., Mahlke, H., & Rosner, J. L. (2008). Quarkonia and their transitions. *Reviews of Modern Physics*, *80*(3), 1161.

Eichten, E. J. (1980). Gottfried K., Kinoshita, T., Lane, K. D., & Yan, T. M. *Phys. Rev. D*, *21*, 203.

Eichten, E. J., & Quigg, C. (1994). Mesons with beauty and charm: Spectroscopy. *Physical Review D*, *49*(11), 5845.

Faustov, R. N., & Galkin, V. O. (2003). Properties of heavy quarkonia and [Formula Presented] mesons in the relativistic quark model. *Physical Review D - Particles, Fields, Gravitation and Cosmology*, *67*(1), 1–19. https://doi.org/10.1103/PhysRevD.67.014027

Fritzsch, H., Gell-Mann, M., & Leutwyler, H. (1973). Advantages of the color octet gluon picture. *Physics Letters B*, *47*(4), 365-368.

Gershtein, S. S., Kiselev, V. V., Likhoded, A. K., & Tkabladze, A. V. (1995). Physics of Bc-mesons. *Physics-Uspekhi*, *38*(1), 1.

Han, K., Böhringer, T., Franzini, P., Mageras, G., Peterson, D., Rice, E., ... & Vogel, H. (1982). Observation of P-Wave b $\bar{b}$ Bound States. *Physical Review Letters*, *49*(22), 1612.

HPQCD, F. L., Allison, I. F., Davies, C. T., Gray, A., Kronfeld, A. S., Mackenzie, P. B., & Simone, J. N. (2005). Mass of the B c meson in three-flavor lattice QCD. *Physical review letters*, *94*(17), 172001.

Innes, W. R., Appel, J. A., Brown, B. C., Brown, C. N., Ueno, K., Yamanouchi, T., ... & Kephart, R. D. (1977). Observation of structure in the ϒ region. *Physical Review Letters*, *39*(20), 1240.

Kiselev, V. V., Kovalsky, A. E., & Onishchenko, A. I. (2001). Heavy quark potential in the static limit of QCD. *Physical Review D*, *64*(5), 054009.

Klopfenstein, C., Horstkotte, J. E., Lee-Franzini, J., Schamberger, R. D., Sivertz, M., Spencer, L. J., ... & Vogel, H. (1983). Observation of the lowest P-wave b b bound states. *Physical Review Letters*, *51*(3), 160.

Kuchin, S. M., & Maksimenko, N. V. (2013). Theoretical estimations of the spin-averaged mass spectra of heavy quarkonia and Bc mesons. *Universal Journal of Physics and Application*, *1*(3), 295-298.

Kwong, W., & Rosner, J. L. (1991). Masses of new particles containing b quarks. *Physical Review D*, *44*(1), 212.

Lees, J. P., Poireau, V., Prencipe, E., Tisserand, V., Tico, J. G., Grauges, E., ... & Schune, M. H. (2011). Study of radiative bottomonium transitions using converted photons. *Physical Review D*, *84*(7), 072002.

Martin, A. (1980). Heavy Flavours and High Energy Collisions in the 1-100 TeV Range. Phys. Lett. B93,338. edited by A. Ali and L. Cifarelli (Plenum Press, New York,1989), p. 141.

Mizuk, R., Asner, D. M., Bondar, A., Pedlar, T. K., Adachi, I., Aihara, H., ... & Tsuboyama, T. (2012). Evidence for the η_b (2 S) and Observation of h_b (1 P)→ η b (1 S) γ and h_b (2 P)→ η_b1 S) γ. *Physical review letters*, *109*(23), 232002.

Motyka, L., & Zalewski, K. (1998). Mass spectra and leptonic decay widths of heavy quarkonia. *The European Physical Journal C-Particles and Fields*, *4*(1), 107-114.

Peacock, A. W. (1986). *Heavy Quarkonium and QCD* (Doctoral dissertation, Durham University).

Peskin, M. E. (2018). *An introduction to quantum field theory*. CRC press.

Quigg, C., & Rosner, J. L. (1977). Quarkonium level spacings. *Physics Letters B*, *71*(1), 153-157.

Rai, A. K., Patel, B., & Vinodkumar, P. C. (2008). Properties of $Q\bar{Q}$ mesons in nonrelativistic QCD formalism. *Physical Review C*, *78*(5), 055202.

Richardson, J. L. (1979). The heavy quark potential & the ϒ, J/ψ systems. *Physics Letters B*, *82*(2), 272-274.

Vijaya Kumar, K. B., B. Hanumaiah, & S. Pepin. "Meson spectrum in a relativistic harmonic model with instanton-induced interaction." *The European Physical Journal A-Hadrons and Nuclei* 19.2 (2004): 247-250.

Workman et al. (Particle Datta Group). (2022). Prog. Theo. Exp. Phys. 2022, 083C01

7 APPENDIX

APPENDIX A: Code

Below is the code that solve Schrodinger wave equation using the Numerov method. Only power-law potential model's function has been made available in the appendix since the function cut across all the potential, and it just a matter of changing the masses of the charm and bottom, and potential models into the specific model.

```
## Importe packages
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.pyplot import figure
import pandas as pd
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', None)
pd.set_option('display.max_colwidth', None)

## Mass, r_min, r_max and dr(h) for power -law potential model
m_b = 5.174
r_min = 0.0
r_max = 7
h = 0.0001
h_sq = h**2
r = np.arange(r_min, r_max, h)
U = np.zeros(len(r))
V = np.zeros(len(r))
g_r = np.zeros(len(r))

## trial energies
trial_energy = np.array([-0.8742, -0.8743, -0.87423])


## Numerov Algorithm
for j in range(len(trial_energy)):
    for i in range(2, len(r)):
        U[0] = 0
        U[1] = h
        V[0] = (-8.064+(6.898)*(r_min)**0.1)
        V[1] = (-8.064+(6.898)*(r_min + h)**0.1)
        g_r[0] = -2*(m_b/2)*(V[0]-trial_energy[j])
        g_r[1] = -2*(m_b/2)*(V[1]-trial_energy[j])

        V[i] =(-8.064+(6.898)*(r[i]**0.1))
        g_r[i] =-2*(m_b/2)*(V[i]-trial_energy[j])
        U[i] = (2*(1-(5.0/12)*h_sq*g_r[i-1])*U[i-1]-(1+(1.0/12)*h_sq*g_r[i-2])*U[i-2])/(1+(1.0/12)*h_sq*g_r[i])



    ## Plot wavefunction
    plt.plot(r, (U/r),  label = 'Eigen energy {} GeV'.format(trial_energy[j]))

 ## Labelling plots
plt.xlim(0,max(r))
plt.title('E = {}GeV'.format(trial_energy))
plt.axhline(y=0, color='k',linestyle='--')
plt.legend()
plt.xlabel(' r [ $GeV^{-1}$]' )
plt.ylabel('R(r)')
plt.savefig('wavebottomPOWER.png')
plt.show()
```

Normalising the radial wavefunction

For 3D schrodinger equation problem in a symmetric potential, the normalised reduced radial function is

$$U(r) = \frac{\phi(r)}{\sqrt{A}}$$

where A is the normalisation integral

```
norm_U = np.zeros(len(r))
A = 0
for i in range(0, len(r)):
    A = A + np.abs(U[i])**2*h
for i in range(0, len(r)):
        norm_U[i] = U[i]/np.sqrt(A)

plt.plot(r, norm_U/r,  label = 'Normalised')
#plt.plot(r, (U/r), label = 'Unnormalised')
plt.axhline(y=0, color='k',linestyle='-')
plt.xlabel(' r [ $GeV^{-1}$]' )
plt.ylabel('R_Normalised')
plt.legend()
plt.savefig('wavebottomPOWERNORM.png')
plt.show()
```

Radial wavefunction at the origin

The schrodinger wavefunction at the origin is given by

$$\psi(0) = \frac{U'(0)}{\sqrt{4\pi}}$$

$$\therefore |\psi(0)|^2 = \frac{1}{4\pi A}$$

Then,

$$|R(0)|^2 = \frac{1}{A}$$

```
R_sq_bb = 1/(A)
print('The radial function at the origin bottomonium is {:.4f} GeV'.format(R_sq_bb))
```

```
The radial function at the origin bottomonium is 4.4242 GeV
```

Mass of the Bottomonium quark

Using this expression $M_{bb} = 2M_b + E_{ns}$ where the E_{ns} is the binding energy which has been calculated above

```
M_bb = 2 * m_b + trial_energy[2]

print('The mass of the bottomonium is {:.4f} GeV'.format(M_bb))
```

```
The mass of the bottomonium is 9.4738 GeV
```

APPENDIX B: Eichten and Quigg result

Table A: The mass and leptonic predictions of the bottomonium system by Eichten and Quigg (Eichten & Quigg, 1994)

Level	Mass (GeV/c^2)		Leptonic Width (keV)	
	Calculated	Observed[a]	Calculated	Observed[a]
1^1S_0 (η_b)	9.377			
1^3S_1 (Υ)	9.464	9.46032 ± 0.00022	1.71	1.34 ± 0.04
2^3P_0 (χ_{b0})	9.834	9.8598 ± 0.0013		
2^3P_1 (χ_{b1})	9.864	9.8919 ± 0.0007		
2^3P_2 (χ_{b2})	9.886	9.9132 ± 0.0006		
2^1P_1 (h_b)	9.873			
2^1S_0 (η_b')	9.963			
2^3S_1 (Υ')	10.007	10.02330 ± 0.00031	0.76	0.586 ± 0.029
3^3D_1	10.120			
3^3D_2	10.126			
3^3D_3	10.130			
3^1D_2	10.127			
3^3P_0 (χ_{b0})	10.199	10.2320 ± 0.0007		
3^3P_1 (χ_{b1})	10.224	10.2549 ± 0.0006		
3^3P_2 (χ_{b2})	10.242	10.26835 ± 0.00057		
3^1P_1 (h_b)	10.231			
3^1S_0	10.298			
3^3S_1	10.339	10.3553 ± 0.0005	0.55	0.44 ± 0.03
4^1S_0	10.573			
4^3S_1	10.602	10.5800 ± 0.0035		

Table B: The mass and leptonic predictions of the charmonium system by Eichten and Quigg (Eichten & Quigg, 1994)

Level	Mass (GeV/c^2)		Leptonic Width (keV)	
	Calculated	Observed[a]	Calculated	Observed[a]
1^1S_0 (η_c)	2.980	2.9788 ± 0.0019		
1^3S_1 (ψ/J)	3.097	$3.09688 \pm 0.00001 \pm 0.00006$[b]	8.00	4.72 ± 0.35
2^3P_0 (χ_{c0})	3.436	3.4151 ± 0.0010		
2^3P_1 (χ_{c1})	3.486	$3.51053 \pm 0.00004 \pm 0.00012$[b]		
2^3P_2 (χ_{c2})	3.507	$3.55615 \pm 0.00007 \pm 0.00012$[b]		
2^1P_1 (h_c)	3.493	$3.5262 \pm 0.00015 \pm 0.0002$[c]		
2^1S_0 (η_c')	3.608			
2^3S_1 (ψ')	3.686	3.68600 ± 0.00010	3.67	2.14 ± 0.21

Table C: The mass and leptonic predictions of the B_c system by Eichten and Quigg (Eichten & Quigg, 1994)

Level	Calculated Mass	Eichten & Feinberg[a]	Gershteĭn et al.[b]	Chen & Kuang
1^1S_0 (B_c)	6.264	6.243	6.246	6.310
1^3S_1 (B_c^*)	6.337	6.339	6.329	6.355
2^3P_0	6.700	6.697	6.645	6.728
$2\ 1^{+\prime}$	6.736	6.740	6.741	6.760
$2\ 1^+$	6.730	6.719	6.682	6.764
2^3P_2	6.747	6.750	6.760	6.773
2^1S_0	6.856	6.969	6.863	6.890
2^3S_1	6.899	7.022	6.903	6.917
3^3D_1	7.012			
3^3D_2	7.012			
3^3D_3	7.005		(7.008)	
3^1D_2	7.009			
3^3P_0	7.108		7.067	7.134
$3\ 1^{+\prime}$	7.142		7.129	7.159
$3\ 1^+$	7.135		7.099	7.160
3^3P_2	7.153		7.143	7.166
3^1S_0	7.244		(7.327)	
3^3S_1	7.280			
4^1S_0	7.562			
4^3S_1	7.594			

APPENDIX C: Gerstein et al result

Table D. The energy levels of the B_c system, calculated without taking into account relativistic corrections (in GeV) (Gerstein et al,1995)

n	[52]	[55]	[54]	n	[52]	[55]	[54]	n	[52]	[55]	[54]
1S	6.301	6.315	6.344	2P	6.728	6.735	6.763	3D	7.008	7.145	7.030
2S	6.893	7.009	6.910	3P	7.122	—	7.160	4D	7.308	—	7.365
3S	7.237	—	7.024	4P	7.395	—	—	5D	7.532	—	—